

Effect of shrub size on survival

Volume and biomass

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1 Introduction

The size of the shrubs in the experiment is a factor that may determine the magnitude of the effect on survival. We anticipate that larger shrubs will exert a more intense effect—whether facilitative or competitive—than smaller ones. Originally, the sampling design considered shrub volume as the most appropriate descriptor of individual size. However, this entails an implicit assumption: that more voluminous shrubs also possess greater biomass.

During our fieldwork and data exploration, we observed that individuals within the pine forest tend to be more voluminous. Nevertheless, this does not appear to be reflected in heavier individuals, as they typically exhibit fewer, thinner branches, and a sparse crown. Therefore, if we were to assume that a more voluminous individual captures more environmental resources due to greater mass, thereby increasing the intensity of competition, or conversely, that a more voluminous individual enriches the soil with more nutrients (in the case of legumes), thereby increasing the intensity of facilitation, we would be overlooking the variability in crown branch density.

To convert volume measurements into biomass, five individuals per environment and shrub species were selected, with volume measurements similar to those of the shrubs chosen for the experiment but without associated seedlings. These individuals were measured, harvested at ground level,

measured *in situ*, and subsequently, a sample was oven-dried in the laboratory. This methodology allowed us to establish a relationship between volume, fresh weight, and dry weight.

2 Objectives

1. To evaluate the best-fit model for predicting shrub biomass.
2. To evaluate the effect of biomass on seedling survival, after controlling for other variables such as elevation.

In previous reports, elevation appeared to have a detectable yet subtle effect on survival. In this regard, an additional objective is:

3. To assess whether the observed effect of elevation remains significant after incorporating this new variable.

3 Methods

```
# Libraries
library(tidyverse)
library(MuMIn)
library(performance)
library(glmmTMB)
library(car)
library(ggeffects)
library(emmeans)
library(multcomp)
```

3.1 Predict biomass

Data Preparation

We utilized a field dataset (`shrub_biomass.csv`) comprising shrub volume (m³) and dry mass (g), alongside species (microsite) and zone type (pine canopy vs. open area) for each individual. Variables were renamed to a standardized English format, and categories for the variables `microsite` (Gs, Cl, Rc → *Genista scorpius*, *Cistus ladanifer*, *Rosa canina*) and `zone` (Open area, Pine canopy) were recoded for interpretability. Both variables were defined as factors.

```
options(na.action = "na.fail")
# Load data
```

```

df.biomass <- read.csv2(file = "00-data/shrub_biomass.csv")

# source("01-scripts/01.2-surv_par_vol.R")

df.biomass <-
df.biomass |>
  rename(volume = Volumen.arbusto..m3.,
         mass = Peso.seco.arbusto..g.,
         log.volume = Log.Volumen,
         log.mass = Log.masa,
         zone = ambiente,
         microsite = Especie) |>
  dplyr::mutate(
    microsite = dplyr::recode(microsite,
                             "Gs" = "Genista scorpius",
                             "Cl" = "Cistus ladanifer",
                             "Rc" = "Rosa canina"
                             ),
    zone = dplyr::recode(zone,
                         "Abierto" = "open area",
                         "pinar" = "pine canopy")
  ) |>
  dplyr::mutate(
    zone = factor(zone),
    microsite = factor(microsite)
  )

head(df.biomass)

```

	ID	microsite	zone	volume	mass	log.volume	log.mass
1	A1	Genista scorpius	open area	0.5597	1584.6	-0.2520511	3.199910
2	A2	Genista scorpius	open area	0.3372	1359.5	-0.4721482	3.133366
3	A3	Genista scorpius	open area	1.5579	3785.8	0.1925275	3.578161
4	A4	Genista scorpius	open area	0.4226	1574.7	-0.3741053	3.197185
5	A5	Genista scorpius	open area	1.8766	6253.8	0.2733676	3.796142
6	J1	Cistus ladanifer	open area	0.6409	631.0	-0.1932189	2.800007

Model Selection

A generalized linear mixed model (`glmmTMB`) was fitted to the log-transformed allometric relationship between mass and volume ($\text{log.mass} \sim \text{log.volume} * \text{microsite} * \text{zone}$), including all pos-

sible interactions between volume, microsite, and zone. From this saturated model, we performed model selection using the `dredge` function from the `MuMIn` package (Bartoń 2023), generating all possible combinations of predictors and selecting the best-fit model (lowest AICc, `subset = 1`). Heteroscedasticity of residuals was detected in the chosen model; therefore, a Gamma distribution was selected to mitigate this issue.¹

```
m_full <- glmmTMB(log.mass ~ log.volume * microsite * zone,
  data = df.biomass)
```

```
dd <- dredge(m_full)
modelo <- get.models(dd, subset = 1)[[1]]
```

```
car::Anova(modelo)
```

Analysis of Deviance Table (Type II Wald chisquare tests)

Response: log.mass

	Chisq	Df	Pr(>Chisq)
log.volume	97.928	1	< 2.2e-16 ***
microsite	68.094	2	1.636e-15 ***
zone	69.936	1	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
parameters::parameters(modelo)
```

Fixed Effects

Parameter	Coefficient	SE	95% CI	z	p
(Intercept)	3.20	0.04	[3.11, 3.28]	71.47	< .001
log volume	0.98	0.10	[0.78, 1.17]	9.90	< .001
microsite [Genista scorpius]	0.28	0.06	[0.17, 0.39]	5.02	< .001
microsite [Rosa canina]	-0.28	0.06	[-0.41, -0.16]	-4.40	< .001
zone [pine canopy]	-0.40	0.05	[-0.49, -0.31]	-8.36	< .001

Dispersion

¹Tengo dudas todavía de si el gamma es el mejor modelo, porque parece aumentar un poco el error.

Parameter	Coefficient	95% CI
(Intercept)	0.12	[0.09, 0.16]

```
performance::performance(modelo)
```

```
# Indices of model performance
```

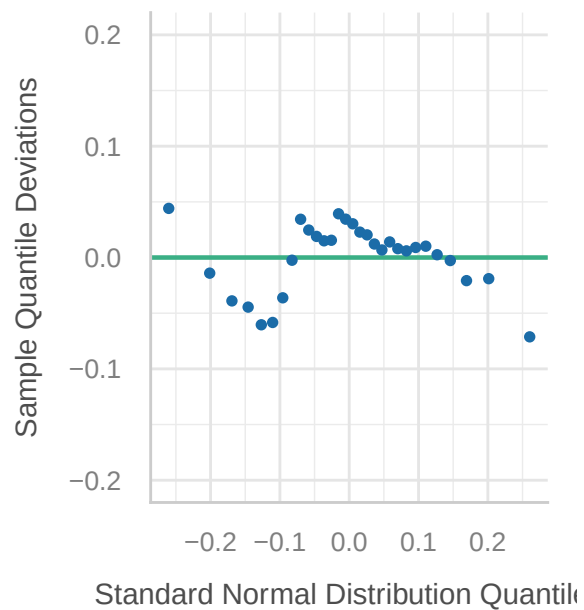
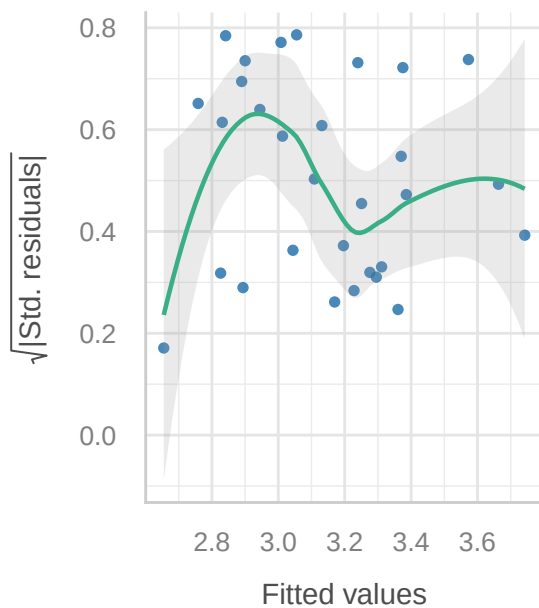
AIC	AICc	BIC	R2	R2 (adj.)	RMSE	Sigma
-28.9	-25.3	-20.5	0.822	0.785	0.122	0.122

```
check_model(modelo, check = c("qq", "homogeneity"))
```

Homogeneity of Variance

Normality of Residuals

Reference line should be flat and horizontal Dots should fall along the line



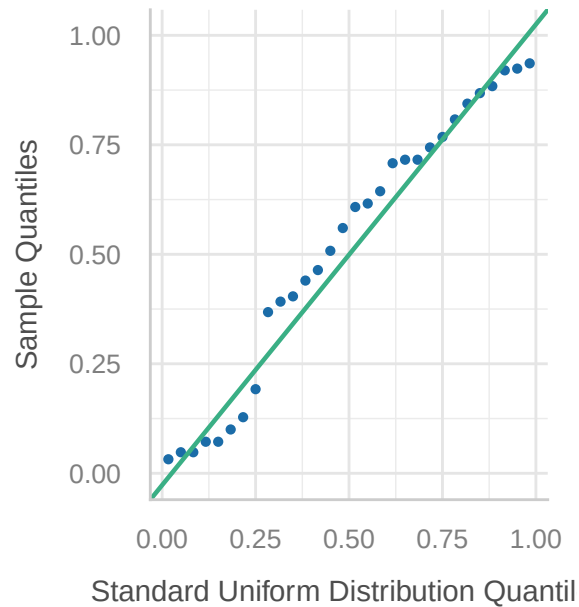
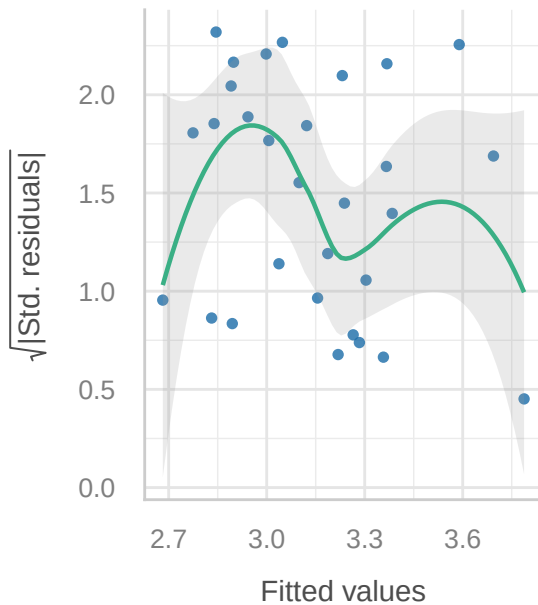
```
m_gamma <- glmmTMB(log.mass ~ log.volume + microsite + zone,
  data = df.biomass, family = Gamma(link = "log"))
```

```
check_model(m_gamma, check = c("qq", "homogeneity"))
```

Homogeneity of Variance

Distribution of Quantile Residuals

Reference line should be flat and horizontal
Dots should fall along the line



```
check_overdispersion(m_gamma)
```

```
# Overdispersion test
```

```
dispersion ratio = 0.988  
p-value = 0.944
```

Treatment-Specific Models

Following the selection of the best model, data were grouped by the `microsite` $\times$ `zone` combination. For each group, a generalized linear model (GLM) was fitted with a Gamma distribution and a log-link function (`mass ~ volume, family = Gamma(link = "log")`). This approach captures the non-linear allometric relationship between volume and mass specifically for each species-environment combination, while accounting for the continuous and strictly positive nature of the response variable (mass). Fitting specific models enhances individual predictive performance. Coefficients (`broom::tidy`) and goodness-of-fit statistics (`broom::glance`) were extracted for each individual model.

```
results_biomass <- df.biomass |>  
  group_by(microsite, zone) |>
```

```

nest() |>                                     # crea list-column
  ↪ 'data'
mutate(
  model = map(data, ~ glm(mass ~ volume, data = .x, family = Gamma(link =
    ↪ "log"))),
  tidied = map(model, broom::tidy),           # coeficientes
  glanced = map(model, broom::glance)        # R2, AIC, etc.
)

```

Biomass Prediction

Survival and shrub volume data were loaded.

```

# Merge predicted biomass
# Cargar supervivencia
df.surv1 <- read.csv2("00-data/vol_par_surv.csv")
df.surv2 <- read.csv2("00-data/sup_growth_may2026.csv")

# Prepare survival for 2025
df.surv1 <- df.surv1 |>
  dplyr::select(micrositio, ambiente, especie_facilitadora,
    ↪ especie_facilitada, # Categorical var
                supervivencia, fecha_supervivencia # survival, response
  ) |>
  dplyr::rename(
    Individual = micrositio,
    Environment = ambiente,
    microsite = especie_facilitadora,
    quercus_sp = especie_facilitada,
    survival = supervivencia,
    surv_date = fecha_supervivencia
  ) |>
  mutate(microsite = dplyr::recode(microsite,
    "claro" = "open",
    "Cistus sp." = "Cistus ladanifer"),
    Environment = dplyr::recode(Environment,
    "abierto" = "gap",
    "pinar" = "pine canopy")
  ) |> unique()

```

```

# Prepare survival for 2026

df.surv2 <- df.surv2 |>
  # Select variables of interest in proper order
  dplyr::select(Arbusto, Ambiente, Especie.arbusto, Quercinea,
                Supervivencia, Fecha) |>
  # Rename variables
  dplyr::rename(
    Individual = Arbusto,
    Environment = Ambiente,
    microsite = Especie.arbusto,
    quercus_sp = Quercinea,
    survival = Supervivencia,
    surv_date = Fecha
  ) # Erase "0" in some individual names (A01, A02...)
  |>
  mutate(
    Individual = sub("([ACJR])0+([1-9][0-9]*)$", "\\1\\2", Individual),
    microsite = dplyr::recode(microsite,
                             "claro" = "open",
                             "Cistus sp." = "Cistus ladanifer"),
    Environment = dplyr::recode(Environment,
                                 "abierto" = "gap",
                                 "pinar" = "pine canopy")
  ) |> unique()

df.surv <- rbind(df.surv1, df.surv2)

head(df.surv)

```

	Individual	Environment	microsite	quercus_sp	survival	surv_date
1	A1	gap	Genista scorpius	QI	1	12/06/2025
2	A1	gap	Genista scorpius	QF	1	12/06/2025
3	A1	gap	Genista scorpius	QI	2	16/09/2025
4	A1	gap	Genista scorpius	QF	0	16/09/2025
5	A10	gap	Genista scorpius	QI	1	12/06/2025
6	A10	gap	Genista scorpius	QF	1	12/06/2025


```

df.vol <- read.csv2("00-data/vol_par_surv.csv")

df.vol <- df.vol |>
  dplyr::select(micrositio, especie_facilitadora, ambiente,
                altura, diametro_1, diametro_2) |>
  mutate(
    volume = 4/3 * pi * altura/2 * diametro_1/2 * diametro_2/2,
    log.volume = log(volume)
  ) |>
  dplyr::rename(
    Individual = micrositio,
    microsite = especie_facilitadora,
    zone = ambiente,
    height = altura,
    diam_1 = diametro_1,
    diam_2 = diametro_2,
  ) |>
  mutate(microsite = dplyr::recode(microsite, "claro" = "gap", "Cistus sp." =
    ↪ "Cistus ladanifer"),
    zone = dplyr::recode(zone, "abierto" = "open area", "pinar" =
    ↪ "pine canopy")
  ) |>
  unique()

head(df.vol)

```

	Individual	microsite	zone	height	diam_1	diam_2	volume
1	A1	Genista scorpius	open area	1.90	1.35	1.05	1.4101824
5	A10	Genista scorpius	open area	0.95	1.30	1.05	0.6789767
9	A11	Genista scorpius	open area	1.35	1.55	1.10	1.2051935
13	A12	Genista scorpius	open area	0.85	1.20	1.00	0.5340708
17	A13	Genista scorpius	open area	1.10	1.14	0.87	0.5712358
21	A14	Genista scorpius	open area	1.76	1.20	0.90	0.9952566
				log.volume			
1				0.343719059			
5				-0.387168449			
9				0.186640120			
13				-0.627226956			
17				-0.559953209			
21				-0.004754733			

The group-specific Gamma models were applied to predict shrub biomass within the original experimental dataset. Individual volumes (excluding the “gap” category) were used, matching each individual with the model corresponding to its specific microsite and zone combination.

```
# Predict biomass values
df.model <-
df.vol |>
  dplyr::select(Individual, volume, microsite, zone) |>
  dplyr::filter(microsite != "gap") |>
  droplevels() |>
  drop_na() |>
  unique()

predictions <- df.model |>
  group_by(microsite, zone) |>
  nest() |>
  left_join(
    results_biomass |>
      dplyr::select(microsite, zone, model),
    by = c("microsite", "zone")
  ) |>
  mutate(
    pred.mass = map2(model, data, ~ predict(.x, newdata = .y, type =
      ↪ "response"))
  ) |>
  unnest(c(data, pred.mass))      # devuelve un dataframe plano
```

Predictive Model Validation

Predictive performance was evaluated by applying the group-specific models to the calibration dataset itself (measured biomass), comparing predicted versus observed values. Two diagnostic plots were generated: (i) measured mass vs. estimated mass to assess general predictive capacity, and (ii) residuals vs. estimated mass to assess homoscedasticity. Model performance was quantified using:

- **RMSE** (Root Mean Square Error)
- **Predictive R^2** , calculated as $1 - (\text{Residual SS} / \text{Total SS})$

```
# Prediction validation
predictions.validation <- df.biomass |>
  group_by(microsite, zone) |>
```

```

nest() |>
left_join(
  results_biomass |> dplyr::select(microsite, zone, model),
  by = c("microsite", "zone")
) |>
mutate(
  pred.mass = map2(model, data, ~ predict(.x, newdata = .y, type =
    ↪ "response"))
) |>
unnest(c(data, pred.mass))      # devuelve un dataframe plano

predictions.validation <- predictions.validation |>
  mutate(
    res = pred.mass - mass
  )

# 1. Configurar la ventana para tener 1 fila y 2 columnas
par(mfrow = c(1, 2))

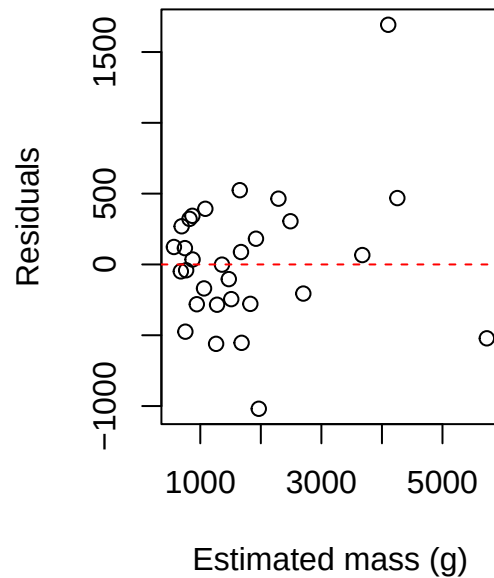
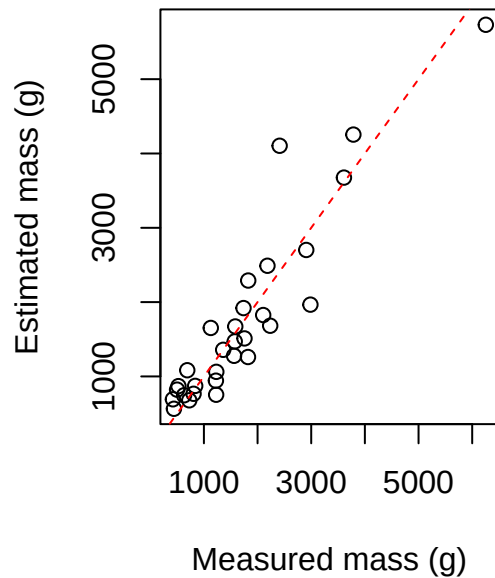
# 2. Primer gráfico (Capacidad predictiva)
plot(
  predictions.validation$mass,
  predictions.validation$pred.mass,
  main = "Predictive capacity of shrub mass.",
  xlab = "Measured mass (g)",
  ylab = "Estimated mass (g)"
)
abline(0, 1, col = "red", lty = 2)

# 3. Segundo gráfico (Homocedasticidad)
plot(
  x = predictions.validation$pred.mass,
  y = predictions.validation$res,
  main = "Homoscedasticity of residuals",
  xlab = "Estimated mass (g)",
  ylab = "Residuals"
)
abline(h = 0, col = "red", lty = 2)

```

Predictive capacity of shrub ma

Homoscedasticity of residual



```
# 4. (Opcional) Restaurar la ventana gráfica a su estado normal (1 fila, 1
  ↪ columna)
par(mfrow = c(1, 1))
```

```
#RMSE
```

```
rmse <- sqrt(mean((predictions.validation$mass -
  ↪ predictions.validation$pred.mass)^2))
```

```
# R2 predictivo
```

```
predictive.r2 <- 1 - sum((predictions.validation$mass -
  ↪ predictions.validation$pred.mass)^2) /
  sum((predictions.validation$mass - mean(predictions.validation$mass))^2)
```

```
cat(
  "Biomass estimation script has run successfully. \n",
  "Predictive capacity and homoscedasticity plots are displayed. \n",
  "Root Mean Square Error (RMSE):", rmse, "\n",
  "Predictive R2:", predictive.r2, "\n"
)
```

Biomass estimation script has run successfully.

Predictive capacity and homoscedasticity plots are displayed.

Root Mean Square Error (RMSE): 473.6756

Predictive R2: 0.8499924

Results Integration

Estimated biomass values were incorporated into the main dataset (`df.surv`), making them available for subsequent analyses. The original shrub dimension data were also retained. Elevation was integrated too.

```
df.surv.mass <- df.surv |>
  left_join(predictions |> dplyr::select(Individual, pred.mass),
            by = "Individual") |>
  dplyr::select(-microsite.y) |>
  rename(
    "microsite" = "microsite.x"
  )

df.surv.vol.mass <-
  merge(x = df.surv.mass,
        y = df.vol,
        by = "Individual")

# Load digital terrain model 5m
r <- terra::rast("00-data/PNOA_MDT05_ETRS89_HU30_0486_LID.tif")

# Load shrub positions
df.shrubs.gps <- read.csv2(file = "00-data/shrubs_gps.csv")

## Change variable names and
## codification of zone
df.shrubs.gps <- df.shrubs.gps |>
  dplyr::select(-X.1, -RP) |>
  rename(
    Individual = Tratamiento,
    Zone       = Ambiente,
    microsite_code = SP
  ) |>
  mutate(
    Zone = dplyr::recode(
      Zone,
      "abierto" = "gap",
```

```

      "pinar " = "pine canopy"
    )
  )

pts <- sf::st_as_sf(
  df.shrubs.gps,
  coords = c("X", "Y"),
  crs = 25830
)

# Convert points to SpatVector
vpts <- terra::vect(pts)

dem_crop <- terra::crop(
  r,
  terra::ext(vpts) + 25
)

dem_mask <- terra::mask(dem_crop, vpts)

# Compute slope
slope <- terra::terrain(
  dem_crop,
  v = "slope",
  unit = "degrees"
)

# extract elevation and slope data
elev <- terra::extract(dem_crop, vpts)
slop <- terra::extract(slope, vpts)
df.shrubs.gps$Elevation <- elev[,2]
df.shrubs.gps$Slope <- slop[,2]

df.surv.vol.mass.elev <- merge(x = df.surv.vol.mass,
  y = df.shrubs.gps |> dplyr::select(Individual, Elevation, Slope),
  by = "Individual")

head(df.surv.vol.mass.elev)

Individual Environment      microsite.x quercus_sp survival  surv_date

```

1	A1	gap	Genista scorpius	QF	1	12/06/2025
2	A1	gap	Genista scorpius	QI	2	16/09/2025
3	A1	gap	Genista scorpius	QF	0	16/09/2025
4	A1	gap	Genista scorpius	QI	1	12/06/2025
5	A1	gap	Genista scorpius	QF	0	27/05/2026
6	A1	gap	Genista scorpius	QI	1	27/05/2026

	zone.x	pred.mass	microsite.y	zone.y	height	diam_1	diam_2	volume
1	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182
2	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182
3	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182
4	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182
5	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182
6	open area	3705.335	Genista scorpius	open area	1.9	1.35	1.05	1.410182

	log.volume	Elevation	Slope
1	0.3437191	1044.26	5.35219
2	0.3437191	1044.26	5.35219
3	0.3437191	1044.26	5.35219
4	0.3437191	1044.26	5.35219
5	0.3437191	1044.26	5.35219
6	0.3437191	1044.26	5.35219

3.2 Model selection

Once the predicted biomass variable was obtained, we proceeded to select the best-fit model to explain survival as a function of treatments, elevation, and shrub biomass. We began with a saturated model; however, instead of **dredge**, we employed a stepwise approach.

Stepwise model selection involves starting with a full model and evaluating the change in AIC upon the removal of higher-order coefficients. If the removal of a coefficient results in a decrease of more than two AIC units, that coefficient is removed, and the process restarts. If no removal results in a decrease of more than two units, the coefficient that yields the greatest AIC reduction is removed. If none of the removals reduce the AIC—but rather increase it—the coefficient that worsens the AIC by less than two units is removed, if applicable. If removing any coefficient increases the AIC by more than two units, the model is considered selected.

The stepwise procedure is an alternative to **dredge**, and both are appropriate for model selection. However, **dredge** evaluates all possible model combinations. In models with many interactions, processing time becomes unmanageable. This is the case with our model, which includes five interactions. While the stepwise procedure is computationally less expensive, it carries a higher risk of falling into a local AIC minimum, potentially precluding the evaluation of better models.

```

df.surv.vol.mass.elev.t3 <-
  df.surv.vol.mass.elev |> filter(surv_date == "27/05/2026")

df <- df.surv.vol.mass.elev.t3 |>
  dplyr::select(survival, Individual, microsite.x, Environment, quercus_sp,
    ↪ Elevation, pred.mass) |>
  drop_na() |>
  rename(
    microsite = microsite.x
  ) |>
  mutate(
    Elevation_c = scale(Elevation),
    log.pred.mass = log(pred.mass),
    log.pred.mass_c = scale(log.pred.mass)
  )

# Modelo saturado
m_full <- glmmTMB(survival ~
  ↪ Environment*microsite*quercus_sp*Elevation_c*log.pred.mass_c +
    (1 | Individual),
  data = df,
  family = binomial()
)

# Iteración manual
m2 <- update(m_full, . ~ . -
  ↪ Environment:microsite:quercus_sp:Elevation_c:log.pred.mass_c)

m3 <- update(m2, . ~ . - Environment:microsite:Elevation_c:log.pred.mass_c)

m4 <- update(m3, . ~ . - Environment:quercus_sp:Elevation_c:log.pred.mass_c)

m5 <- update(m4, . ~ . - Environment:microsite:quercus_sp:log.pred.mass_c)

m6 <- update(m5, . ~ . - Environment:microsite:log.pred.mass_c)

m7 <- update(m6, . ~ . - Environment:Elevation_c:log.pred.mass_c)

m8 <- update(m7, . ~ . - microsite:quercus_sp:Elevation_c:log.pred.mass_c)

```



```

m9 <- update(m8, . ~ . - quercus_sp:Elevation_c:log.pred.mass_c)

m10 <- update(m9, . ~ . - microsite:quercus_sp:log.pred.mass_c)

m11 <- update(m10, . ~ . - microsite:Elevation_c:log.pred.mass_c)

m12 <- update(m11, . ~ . - Elevation_c:log.pred.mass_c)

m13 <- update(m12, . ~ . - Environment:quercus_sp:log.pred.mass_c)

m14 <- update(m13, . ~ . - quercus_sp:log.pred.mass_c)

# Hay una interaccion de 4v sin interacciones significativas de 3 vias por
↳ debajo, eliminarla y comparar explicitamente
m15 <- update(m14, . ~ . - Environment:microsite:quercus_sp:Elevation_c)
# AIC(m14, m15)

m16 <- update(m15, . ~ . - microsite:quercus_sp:Elevation_c)

m17 <- update(m16, . ~ . - Environment:microsite:quercus_sp)

m18 <- update(m17, . ~ . - Environment:microsite:Elevation_c)

# Eliminar primero el que más baja el AIC
m19 <- update(m18, . ~ . - microsite:quercus_sp)

m20 <- update(m19, . ~ . - Environment:quercus_sp:Elevation_c)

m21 <- update(m20, . ~ . - quercus_sp:Elevation_c)

m22 <- update(m21, . ~ . - Environment:quercus_sp)

m23 <- update(m22, . ~ . - Environment:log.pred.mass_c)

m24 <- update(m23, . ~ . - Environment:Elevation_c)

m25 <- update(m24, . ~ . - microsite:Elevation_c)

m26 <- update(m25, . ~ . - Elevation_c)

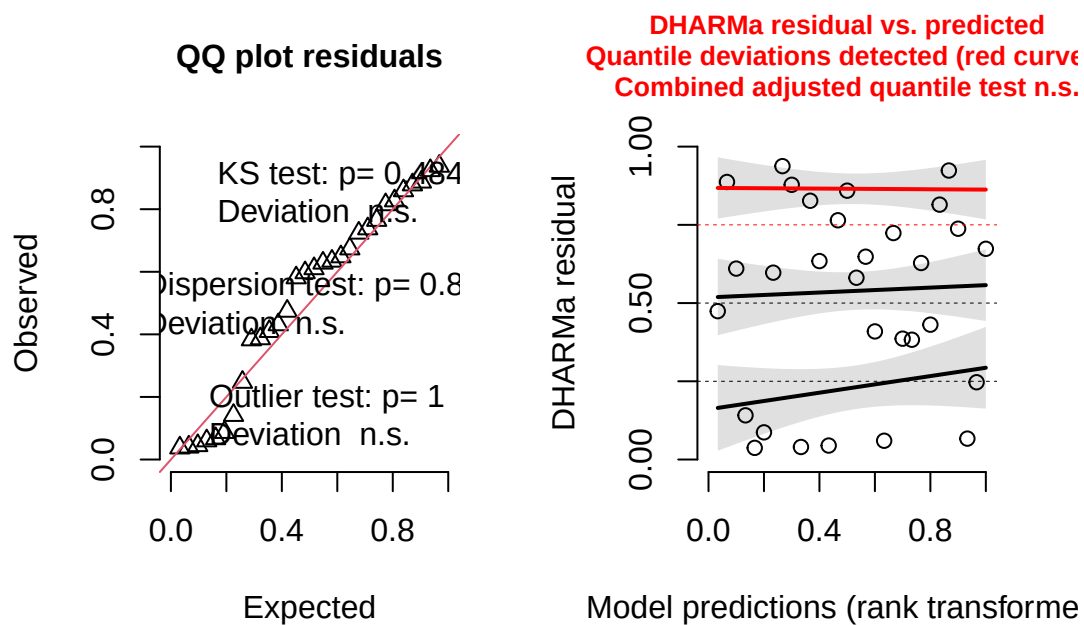
m26a <- update(m26, . ~ . - Environment:microsite)

```

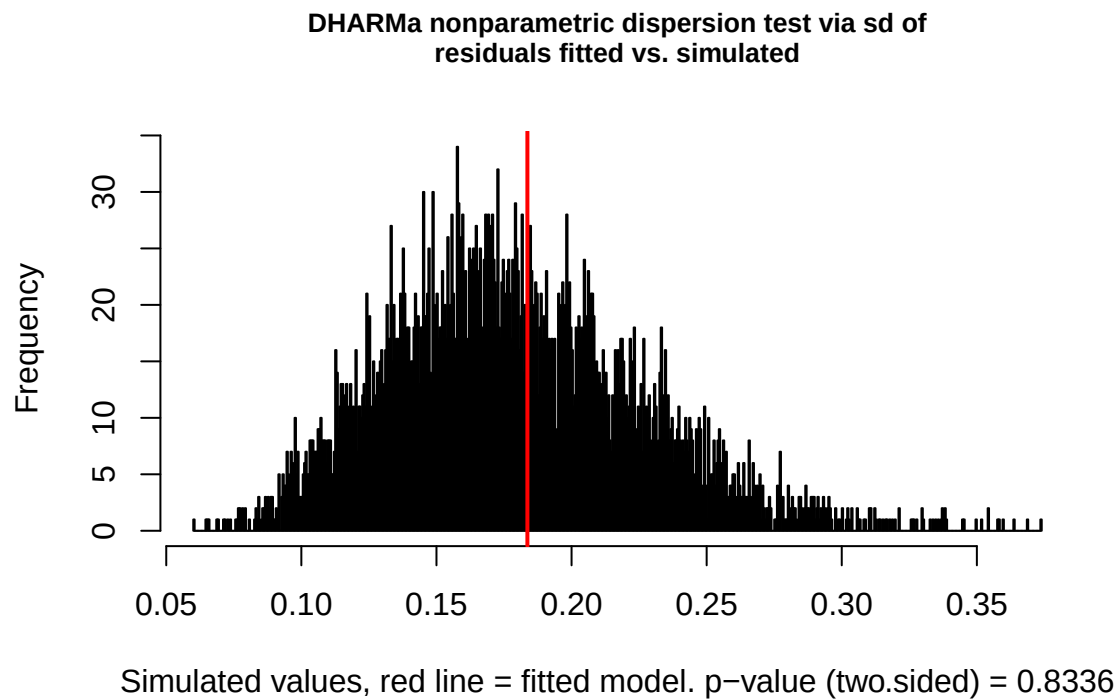
```
m26b <- update(m26, . ~ . - quercus_sp)
m26c <- update(m26, . ~ . - microsite:log.pred.mass_c)
```

```
# Residuos con DHARMA (diseñado para glmmTMB)
library(DHARMA)
sim_res <- simulateResiduals(modelo, n = 5000)
plot(sim_res)
```

DHARMA residual



```
testDispersion(sim_res)
```



DHARMa nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput
dispersion = 1.0304, p-value = 0.8336
alternative hypothesis: two.sided
```

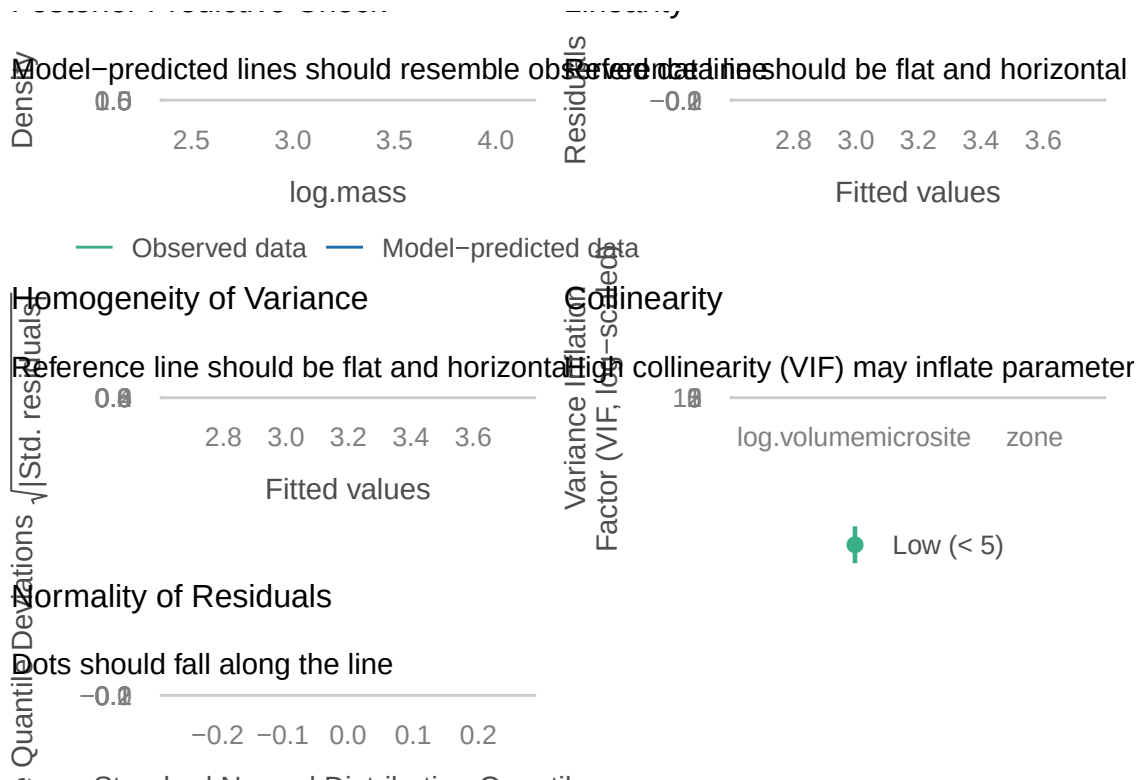
```
library(performance)
```

```
performance(modelo)
```

```
# Indices of model performance
```

AIC	AICc	BIC	R2	R2 (adj.)	RMSE	Sigma
-28.9	-25.3	-20.5	0.822	0.785	0.122	0.122

```
check_model(modelo)
```



```
check_collinearity(modelo)
```

```
# Check for Multicollinearity
```

Low Correlation

Term	VIF	VIF 95% CI	adj. VIF	Tolerance	Tolerance 95% CI
log.volume	1.81	[1.35, 2.88]	1.34	0.55	[0.35, 0.74]
microsite	1.66	[1.26, 2.65]	1.13	0.60	[0.38, 0.79]
zone	1.15	[1.02, 2.45]	1.07	0.87	[0.41, 0.98]

4 Results

Stepwise Procedure

Model selection was performed using a backward stepwise elimination procedure based on the Akaike Information Criterion (AIC). We started with a saturated model that included all main effects of **Environment**, **microsite**, **quercus_sp**, **Elevation_c**, and **log.pred.mass_c**, as well as all possible interactions between them, in addition to a random intercept for the individual.

During this process, there was a nearly continuous reduction in AIC, from **469.61** in the saturated model to **425.24** in the final model ($\Delta\text{AIC} = 44.37$), indicating that the simplification substantially improved the trade-off between model fit and complexity.

The simplification followed a logical hierarchy, first removing higher-order interactions. Initially, five-way interactions involving different combinations of **Environment**, **microsite**, **quercus_sp**, **Elevation_c**, and **log.pred.mass_c** were successively removed. Subsequently, remaining four-way interactions were removed, followed by various three-way interactions that were no longer supported by the data. Once higher-order interactions were discarded, double interactions were simplified, successively eliminating **quercus_sp** $\times$ **Elevation_c**, **Environment** $\times$ **quercus_sp**, **Environment** $\times$ **log.pred.mass_c**, **Environment** $\times$ **Elevation_c**, and **microsite** $\times$ **Elevation_c**.

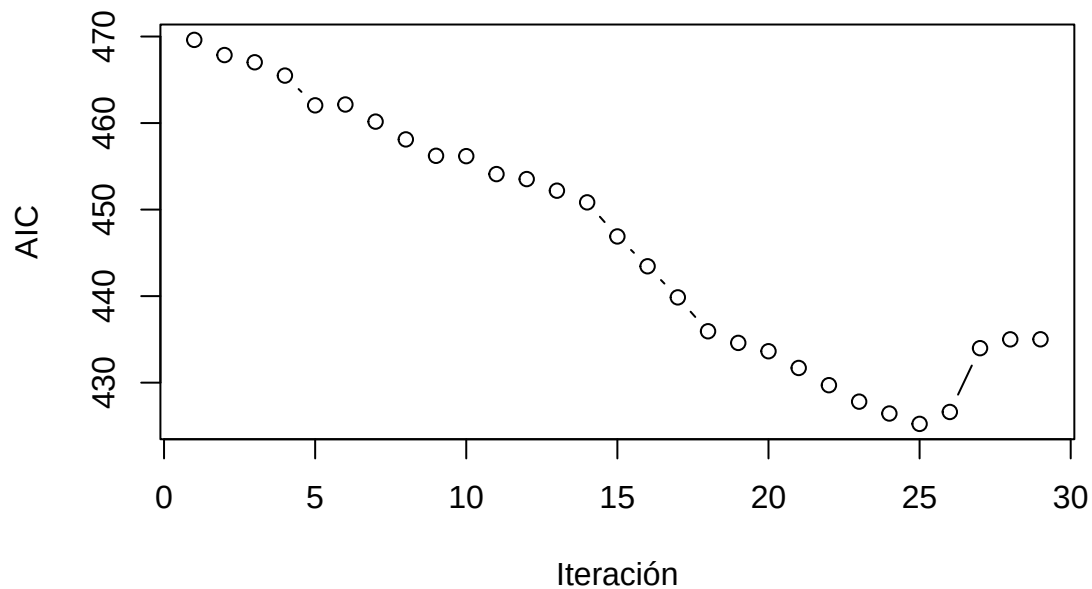
The last effect removed was the main effect of **Elevation_c**. Although its removal produced a slight increase in AIC (425.24 versus 426.61; $\Delta\text{AIC} = 1.38$), this increase was below the two-unit threshold established during the selection procedure; therefore, we opted for a more parsimonious model without this predictor. This result is particularly interesting because, in previous analyses, elevation showed a subtle but significant effect on survival. However, after incorporating predicted shrub biomass (**log.pred.mass_c**) into the set of predictors, the independent contribution of elevation no longer justified its retention in the model, suggesting that part of the pattern previously attributed to elevation might be explained by differences in shrub size.

The final model retained four main effects (**Environment**, **microsite**, **quercus_sp**, and **log.pred.mass_c**) and two double interactions (**Environment** $\times$ **microsite** and **microsite** $\times$ **log.pred.mass_c**). According to Type II Wald tests, **Environment** had a highly significant effect on survival ($P < 0.001$), as did **quercus_sp** ($P = 0.0026$). The main effect of **microsite** was not significant ($P = 0.55$), but it participated in two significant interactions, indicating that its effect depends on both the environment and shrub size. Specifically, the **Environment** $\times$ **microsite** ($P = 0.014$) and **microsite** $\times$ **log.pred.mass_c** ($P = 0.016$) interactions remained in the final model, while the main effect of **log.pred.mass_c** was not significant ($P = 0.376$). This pattern indicates that shrub biomass does not exert a uniform effect on survival; rather, its influence depends on the microsite in which the seedling is established.

```
# Graficar proceso selectivo
mods <- list(
  m_full, m2, m3, m4, m5, m6, m7, m8, m9, m10, m11,
  m12, m13, m14, m15, m16, m17, m18, m19, m20, m21, m22, m23, m24, m25, m26, m26a,
  ↪ m26b, m26c
)

aics <- sapply(mods, AIC)
```

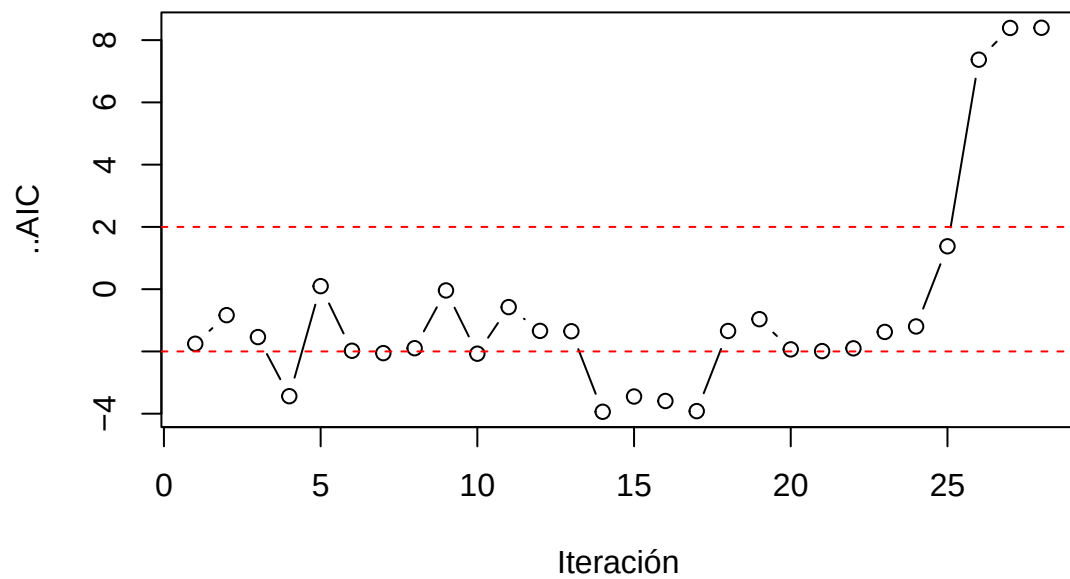
```
plot(aics, type="b",
     xlab="Iteración",
     ylab="AIC")
```



```
# Calculamos los valores de la diferencia
d_aics <- c(diff(aics[1:26]),
            aics[27] - aics[26],
            aics[28] - aics[26],
            aics[29] - aics[26])

# Graficamos ajustando el ylim para que incluya -2 y 2
plot(d_aics, type="b",
     xlab="Iteración",
     ylab="ΔAIC",
     ylim=range(c(d_aics, -2, 2))) # Esto fuerza a que -2 y 2 siempre
    ↪ aparezcan

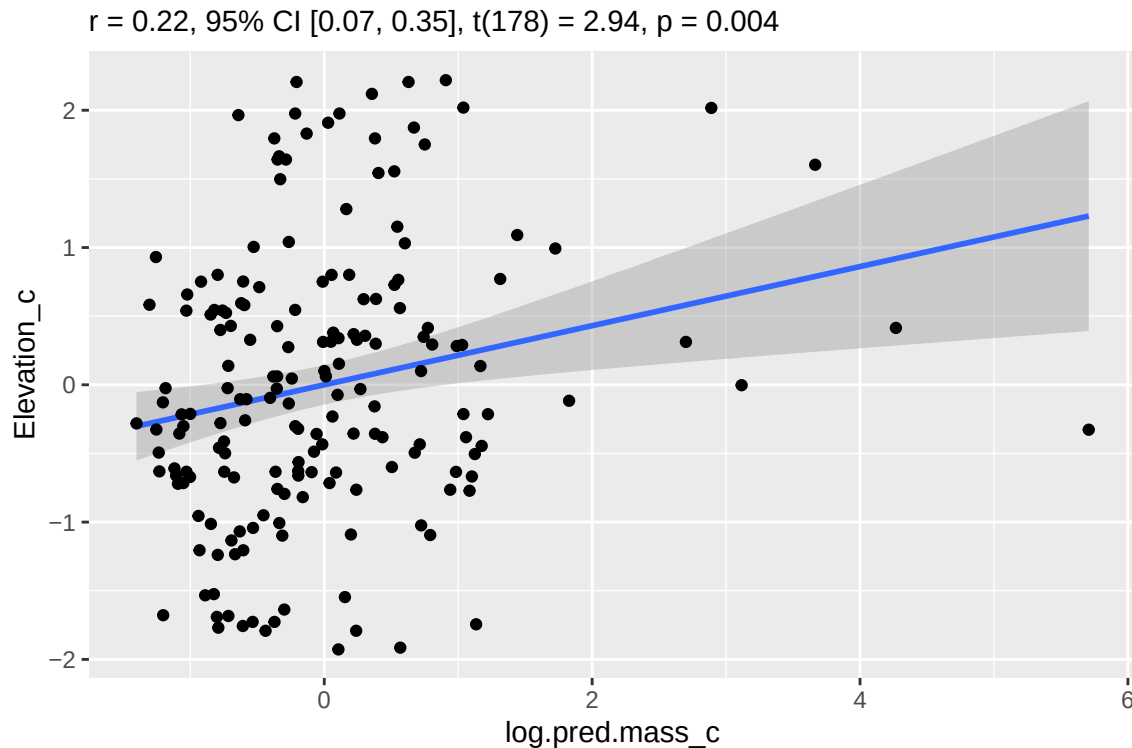
# Ahora añadimos las líneas
abline(h = c(-2, 2), col = "red", lty = 2)
```



The fact that elevation no longer has a significant effect in the model once biomass is introduced suggests a possible high correlation between biomass and elevation. To test this, a Pearson correlation test was applied.

```
df.elev.mass <- df |>
  dplyr::select(Individual, microsite, Environment, Elevation, Elevation_c,
    ↪ pred.mass, log.pred.mass, log.pred.mass_c) |>
  unique()

plot(correlation::cor_test(df.elev.mass, x = "log.pred.mass_c", y =
  ↪ "Elevation_c"))
```



The correlation is **positive but weak** ($r = 0.22$) and statistically significant ($p = 0.004$). This indicates that, on average, shrubs tend to have greater biomass as elevation increases.

Therefore, this result is **consistent** with the hypothesis that part of the effect previously attributed to elevation is captured by shrub biomass when it is incorporated into the model. However, the correlation is too modest to conclude that both variables are equivalent or that there is a significant problem with collinearity. Instead, it suggests a **partial overlap** in the information they provide.

The removal of elevation in the final step suggests that, although its contribution is small, it could function well as a covariate to increase the rigor of our conclusions by allowing us to control for the potential effect of elevation.

4.1 Chosen model

The model results show that survival was primarily determined by **Environment**, the **Quercus species**, and two interactions: **Environment** $\times$ **microsite** and **microsite** $\times$ **shrub biomass** (*log.pred.mass_c*).

```
# Confirmar que es el mejor de toda la secuencia
AIC(m_full, m26)
```

df	AIC
----	-----


```
m_full 49 469.6110
m26     11 426.6138
```

```
# Modelo seleccionado
modelo <- update(m26, REML = TRUE)

# Tabla de efectos
car::Anova(modelo)
```

Analysis of Deviance Table (Type II Wald chisquare tests)

Response: survival

	Chisq	Df	Pr(>Chisq)	
Environment	24.4971	1	7.442e-07	***
microsite	1.1870	2	0.552385	
quercus_sp	9.0555	1	0.002619	**
log.pred.mass_c	0.7840	1	0.375934	
Environment:microsite	8.5166	2	0.014146	*
microsite:log.pred.mass_c	8.3090	2	0.015694	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Resumen completo
summary(modelo)
```

```
Family: binomial (logit)
Formula:
survival ~ Environment + microsite + quercus_sp + log.pred.mass_c +
  (1 | Individual) + Environment:microsite + microsite:log.pred.mass_c
Data: df
```

AIC	BIC	logLik	-2*log(L)	df.resid
430.6	473.4	-204.3	408.6	349

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
Individual	(Intercept)	1.42	1.192

Number of obs: 360, groups: Individual, 180

Conditional model:

	Estimate	Std. Error	z value
(Intercept)	-1.165734	0.407460	-2.861
Environmentpine canopy	0.401673	0.576269	0.697
micrositeGenista scorpius	-1.239029	0.610884	-2.028
micrositeRosa canina	-0.694076	0.574842	-1.207
quercus_spQI	0.775936	0.257851	3.009
log.pred.mass_c	0.003142	0.224727	0.014
Environmentpine canopy:micrositeGenista scorpius	2.379756	0.829909	2.868
Environmentpine canopy:micrositeRosa canina	1.550732	0.795007	1.951
micrositeGenista scorpius:log.pred.mass_c	1.006374	0.439989	2.287
micrositeRosa canina:log.pred.mass_c	-0.860933	0.643956	-1.337

Pr(>|z|)

(Intercept)	0.00422 **
Environmentpine canopy	0.48579
micrositeGenista scorpius	0.04253 *
micrositeRosa canina	0.22727
quercus_spQI	0.00262 **
log.pred.mass_c	0.98884
Environmentpine canopy:micrositeGenista scorpius	0.00414 **
Environmentpine canopy:micrositeRosa canina	0.05111 .
micrositeGenista scorpius:log.pred.mass_c	0.02218 *
micrositeRosa canina:log.pred.mass_c	0.18124

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Parámetros

parameters::parameters(modelo)

Fixed Effects

Parameter	Log-Odds	SE
(Intercept)	-1.17	0.41
Environment [pine canopy]	0.40	0.58
microsite [Genista scorpius]	-1.24	0.61
microsite [Rosa canina]	-0.69	0.57
quercus sp [QI]	0.78	0.26
log pred mass c	3.14e-03	0.22

Environment [pine canopy] × microsite [Genista scorpius]		2.38		0.83
Environment [pine canopy] × microsite [Rosa canina]		1.55		0.80
microsite [Genista scorpius] × log pred mass c		1.01		0.44
microsite [Rosa canina] × log pred mass c		-0.86		0.64

Parameter		95% CI
-----------	--	--------

(Intercept)		[-1.96, -0.37]
Environment [pine canopy]		[-0.73, 1.53]
microsite [Genista scorpius]		[-2.44, -0.04]
microsite [Rosa canina]		[-1.82, 0.43]
quercus sp [QI]		[0.27, 1.28]
log pred mass c		[-0.44, 0.44]
Environment [pine canopy] × microsite [Genista scorpius]		[0.75, 4.01]
Environment [pine canopy] × microsite [Rosa canina]		[-0.01, 3.11]
microsite [Genista scorpius] × log pred mass c		[0.14, 1.87]
microsite [Rosa canina] × log pred mass c		[-2.12, 0.40]

Parameter		z		p
-----------	--	---	--	---

(Intercept)		-2.86		0.004
Environment [pine canopy]		0.70		0.486
microsite [Genista scorpius]		-2.03		0.043
microsite [Rosa canina]		-1.21		0.227
quercus sp [QI]		3.01		0.003
log pred mass c		0.01		0.989
Environment [pine canopy] × microsite [Genista scorpius]		2.87		0.004
Environment [pine canopy] × microsite [Rosa canina]		1.95		0.051
microsite [Genista scorpius] × log pred mass c		2.29		0.022
microsite [Rosa canina] × log pred mass c		-1.34		0.181

Random Effects

Parameter		Coefficient		95% CI
-----------	--	-------------	--	--------

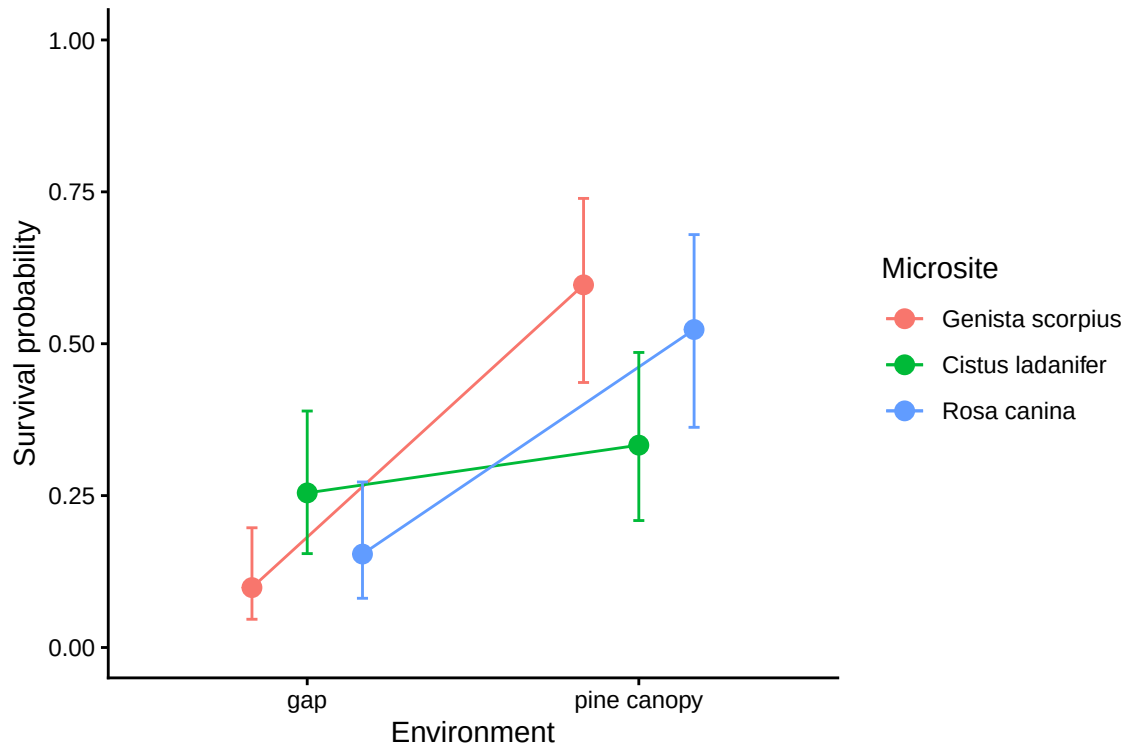
SD (Intercept: Individual)		1.19		[0.69, 2.07]
----------------------------	--	------	--	--------------

The **Environment** × **microsite** interaction reveals that the effect of the microsite depends on the environment in which the seedlings are established. In the **gap**, estimated survival probabilities were 11.7% under *Genista scorpius*, 18.7% under *Rosa canina*, and 31.5% under *Cistus ladanifer*, although these differences were not statistically significant. Conversely, in the **pine canopy**, survival increased to 40.7% under *Cistus ladanifer*, 61.8% under *Rosa canina*, and 68.2% under

Genista scorpius, likewise showing no significant differences between microsites. However, when comparing both environments for each microsite, significant differences emerge for *Genista scorpius* and *Rosa canina*, whose survival values are considerably higher under the pine canopy than in the gaps. Conversely, *Cistus ladanifer* exhibits similar probabilities in both environments, indicating greater stability in its facilitative effect regardless of the environment.

```
pred.env.micro <- ggpredict(
  modelo,
  terms = c("Environment", "microsite"),
  type = "fixed"
)

ggplot(pred.env.micro,
  aes(x = x,
    y = predicted,
    colour = group,
    group = group)) +
  geom_point(size = 3, position = position_dodge(width = .5)) +
  geom_line(position = position_dodge(width = .5)) +
  geom_errorbar(aes(ymin = conf.low,
    ymax = conf.high),
    width = 0.1,
    position = position_dodge(width = .5)) +
  labs(x = "Environment",
    y = "Survival probability",
    colour = "Microsite") +
  ylim(0,1) +
  theme_classic()
```



```
emm_env_micro <- emmeans(
  modelo,
  ~ Environment * microsite,
  type = "response"
)

cld(emm_env_micro, Letters = letters, by = "Environment")
```

Environment = gap:

microsite	prob	SE	df	asympt.LCL	asympt.UCL	.group
Genista scorpius	0.117	0.0506	Inf	0.0486	0.257	a
Rosa canina	0.187	0.0669	Inf	0.0882	0.353	a
Cistus ladanifer	0.315	0.0817	Inf	0.1794	0.491	a

Environment = pine canopy:

microsite	prob	SE	df	asympt.LCL	asympt.UCL	.group
Cistus ladanifer	0.407	0.0959	Inf	0.2396	0.599	a
Rosa canina	0.618	0.0992	Inf	0.4150	0.787	a
Genista scorpius	0.682	0.0865	Inf	0.4957	0.824	a

Results are averaged over the levels of: quercus_sp

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

P value adjustment: tukey method for comparing a family of 3 estimates

Tests are performed on the log odds ratio scale

significance level used: alpha = 0.05

NOTE: If two or more means share the same grouping symbol,
then we cannot show them to be different.

But we also did not show them to be the same.

```
cld(emm_env_micro, Letters = letters, by = "microsite")
```

microsite = *Cistus ladanifer*:

Environment	prob	SE	df	asympt.LCL	asympt.UCL	.group
gap	0.315	0.0817	Inf	0.1794	0.491	a
pine canopy	0.407	0.0959	Inf	0.2396	0.599	a

microsite = *Genista scorpius*:

Environment	prob	SE	df	asympt.LCL	asympt.UCL	.group
gap	0.117	0.0506	Inf	0.0486	0.257	a
pine canopy	0.682	0.0865	Inf	0.4957	0.824	b

microsite = *Rosa canina*:

Environment	prob	SE	df	asympt.LCL	asympt.UCL	.group
gap	0.187	0.0669	Inf	0.0882	0.353	a
pine canopy	0.618	0.0992	Inf	0.4150	0.787	b

Results are averaged over the levels of: *quercus_sp*

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

Tests are performed on the log odds ratio scale

significance level used: alpha = 0.05

NOTE: If two or more means share the same grouping symbol,
then we cannot show them to be different.

But we also did not show them to be the same.

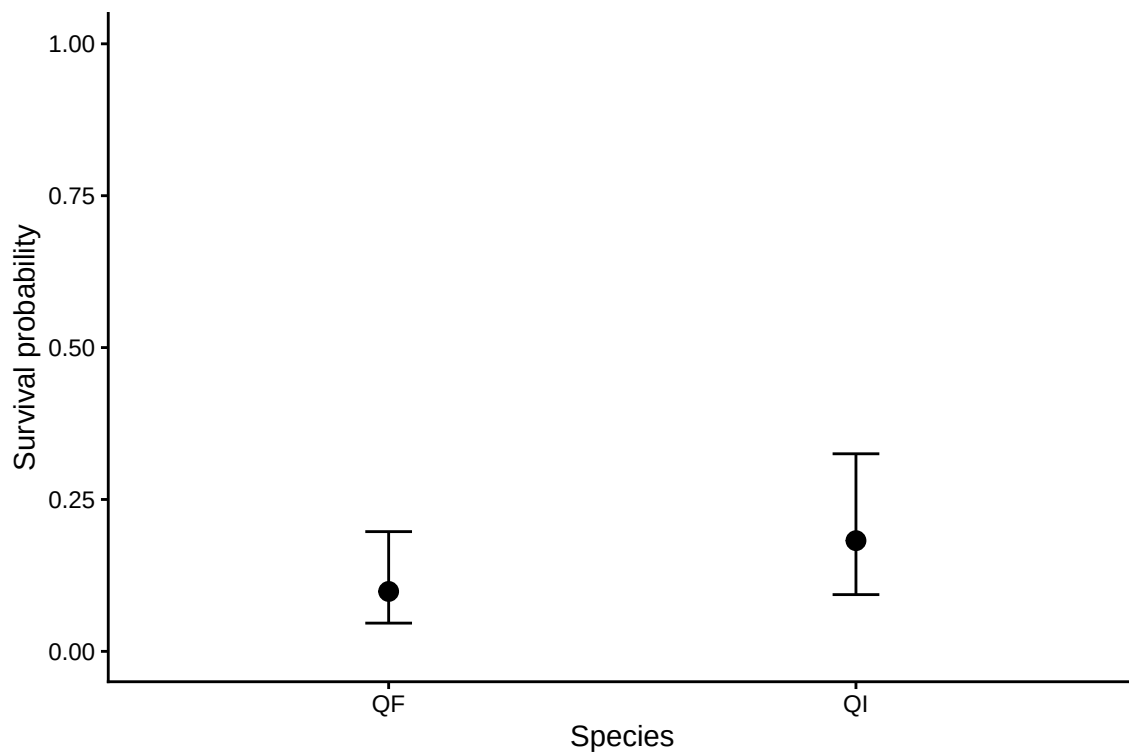
The **Quercus species** also showed a clear effect on survival. Estimated marginal probabilities were 27.8% for **Q. faginea** and 45.6% for **Q. ilex**, a difference that was statistically significant. Therefore, regardless of the environment and microsite, **Q. ilex** exhibits a higher probability of survival.

```

pred.sp <- ggpredict(
  modelo,
  terms = "quercus_sp",
  type = "fixed"
)

ggplot(pred.sp,
  aes(x = x,
      y = predicted)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = conf.low,
                    ymax = conf.high),
                width = 0.1) +
  labs(x = "Species",
       y = "Survival probability") +
  ylim(0,1) +
  theme_classic()

```



```

emm_species <- emmeans(
  modelo,
  ~ quercus_sp,

```

```

    type = "response"
  )

```

```

cld(emm_species, Letters = letters)

```

quercus_sp	prob	SE	df	asympt.LCL	asympt.UCL	.group
QF	0.278	0.0462	Inf	0.197	0.377	a
QI	0.456	0.0527	Inf	0.355	0.559	b

Results are averaged over the levels of: Environment, microsite

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

Tests are performed on the log odds ratio scale

significance level used: alpha = 0.05

NOTE: If two or more means share the same grouping symbol,
 then we cannot show them to be different.

But we also did not show them to be the same.

The interaction between **microsite** and **shrub biomass** reveals a more complex pattern. Estimating survival probabilities for low (5th percentile) and high (95th percentile) biomass values, it is observed that under *Genista scorpius*, survival increases notably with shrub size, rising from approximately 15% to 67%, a significant difference. Under *Cistus ladanifer*, conversely, survival remains practically constant (36% at both biomass extremes), while under *Rosa canina*, the estimate even decreases (from 62% to 16%), although this difference does not reach statistical significance due to the high uncertainty of the estimates.

Comparisons between microsites for the same level of biomass show an apparently different pattern. With low biomass, *Rosa canina* exhibits significantly higher survival than *Genista scorpius*, while *Cistus ladanifer* occupies an intermediate position. However, with high biomass, no significant differences between the three microsites are detected. This result may appear contradictory to the graphical representation, where the positive slope of *Genista scorpius* suggests a clear increase in survival as biomass increases. Nevertheless, both observations are compatible. The interaction reflects that the **survival response to biomass differs among microsites**, whereas marginal mean comparisons evaluate only differences at two specific biomass values. At the upper end of the gradient, the uncertainty of the estimates (wide confidence intervals) makes differences between microsites no longer statistically detectable, even though the visual trend indicates a greater facilitative capacity of *Genista scorpius* when it reaches larger sizes.

```

pred.mass <- ggpredict(
  modelo,
  terms = c("log.pred.mass_c [all]", "microsite"),

```



```

    type = "fixed"
  )

ggplot() +

  ## Observaciones
  geom_jitter(
    data = df,
    aes(x = log.pred.mass_c,
        y = survival,
        colour = microsite),
    width = 0,
    height = 0.03,
    alpha = 0.2,
    size = 2
  ) +

  ## Predicciones
  geom_line(
    data = pred.mass,
    aes(x = x,
        y = predicted,
        colour = group),
    linewidth = 1
  ) +

  ## IC
  geom_ribbon(
    data = pred.mass,
    aes(x = x,
        ymin = conf.low,
        ymax = conf.high,
        fill = group),
    alpha = 0.1,
    colour = NA
  ) +

  labs(
    x = "Predicted shrub biomass (log)",
    y = "Survival probability",
    colour = "Microsite",

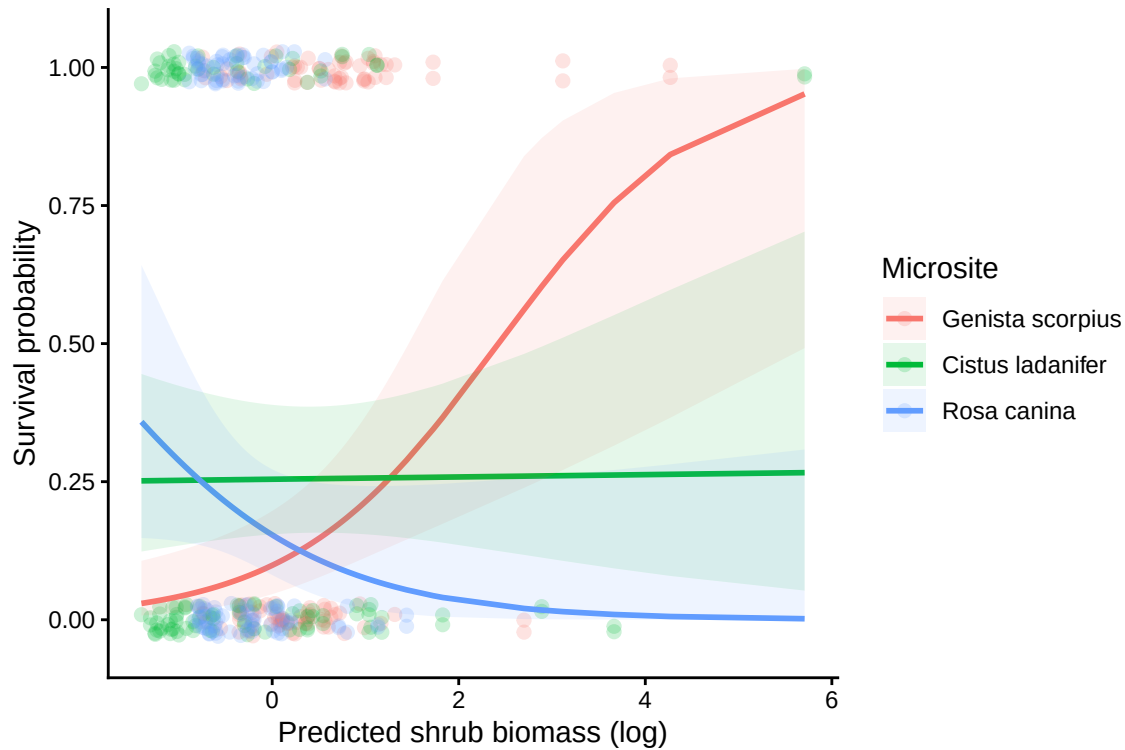
```

```

    fill = "Microsite"
  ) +

  coord_cartesian(ylim = c(-0.05, 1.05)) +
  theme_classic()

```



```

vec <- as.vector(quantile(df.elev.mass$log.pred.mass_c,
  probs = c(.05, .95)))

```

```

emm_mass <- emmeans(
  modelo,
  ~ microsite | log.pred.mass_c,
  at = list(
    log.pred.mass_c = vec
  ),
  type = "response"
)

```

```

cld(emm_mass, Letters = letters, by = "microsite")

```

```

microsite = Cistus ladanifer:

```

log.pred.mass_c	prob	SE	df	asympt.LCL	asympt.UCL	.group
-1.12	0.359	0.0762	Inf	0.2262	0.517	a
1.32	0.361	0.0981	Inf	0.1967	0.565	a

microsite = *Genista scorpius*:

log.pred.mass_c	prob	SE	df	asympt.LCL	asympt.UCL	.group
-1.12	0.147	0.0816	Inf	0.0459	0.381	a
1.32	0.670	0.0984	Inf	0.4588	0.829	b

microsite = *Rosa canina*:

log.pred.mass_c	prob	SE	df	asympt.LCL	asympt.UCL	.group
1.32	0.164	0.1400	Inf	0.0257	0.594	a
-1.12	0.615	0.1320	Inf	0.3489	0.826	a

Results are averaged over the levels of: Environment, *quercus_sp*

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

Tests are performed on the log odds ratio scale

significance level used: alpha = 0.05

NOTE: If two or more means share the same grouping symbol,

then we cannot show them to be different.

But we also did not show them to be the same.

```
cld(emm_mass, Letters = letters, by = "log.pred.mass_c")
```

log.pred.mass_c = -1.12:

microsite	prob	SE	df	asympt.LCL	asympt.UCL	.group
<i>Genista scorpius</i>	0.147	0.0816	Inf	0.0459	0.381	a
<i>Cistus ladanifer</i>	0.359	0.0762	Inf	0.2262	0.517	ab
<i>Rosa canina</i>	0.615	0.1320	Inf	0.3489	0.826	b

log.pred.mass_c = 1.32:

microsite	prob	SE	df	asympt.LCL	asympt.UCL	.group
<i>Rosa canina</i>	0.164	0.1400	Inf	0.0257	0.594	a
<i>Cistus ladanifer</i>	0.361	0.0981	Inf	0.1967	0.565	a
<i>Genista scorpius</i>	0.670	0.0984	Inf	0.4588	0.829	a

Results are averaged over the levels of: Environment, *quercus_sp*

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

P value adjustment: tukey method for comparing a family of 3 estimates

Tests are performed on the log odds ratio scale

significance level used: $\alpha = 0.05$

NOTE: If two or more means share the same grouping symbol,
then we cannot show them to be different.

But we also did not show them to be the same.

Bartoń, Kamil. 2023. *MuMIn: Multi-Model Inference*. <https://CRAN.R-project.org/package=MuMIn>.